

## SMART INDIA HACKATHON 2026

# APIx: National Airfare Price Index & Regulatory Intelligence Platform

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| <b>Problem Statement Title:</b> | Development of an Airfare Price Index (APIx) for Domestic Scheduled Aviation in India |
| <b>Theme:</b>                   | Smart Automation / Transportation & Logistics / Regulatory Technology                 |
| <b>Category:</b>                | Software                                                                              |
| <b>Target Authority:</b>        | Ministry of Civil Aviation (MoCA) & Directorate General of Civil Aviation (DGCA)      |
| <b>Team ID:</b>                 | [Your Team ID]                                                                        |
| <b>Team Name:</b>               | [Your Team Name]                                                                      |

# Slide 2: Proposed Solution & Innovation

## 1. Proposed Solution & Working Prototype

- **80-Route Sovereign Domestic Basket:** Tracks comprehensive fare distributions across 20 metro, secondary, and UDAN airport pairs covering ~92% of scheduled seat capacity.
- **5 Multi-Lead Horizons:** Continuously audits price volatility across T+1 (Emergency/Last-minute), T+7, T+15, T+30, and T+45 booking lead times.
- **Modified Laspeyres Methodology:** Mathematically robust index formula (Base 100.0) weighted by quarterly DGCA Form A/B passenger traffic statistics ( $w_p$ ).
- **Statistical IQR Filtration:** Automatic  $[Q1 - 1.5 \cdot IQR, Q3 + 1.5 \cdot IQR]$  bounds trimming predatory gouging spikes & promotional flash fares before computing medians.

## 2. How it Addresses the Problem & Key Innovations

- **Eliminates Regulatory Blindspots:** Replaces ad-hoc manual spot-checking with systematic, real-time macro and micro airfare indices.
- **1-Click DGCA Executive Brief Generator:** Auto-compiles live anomaly audits & antitrust matrices into a printable A4 Ministry memo.
- **Interactive ATF Shock Simulator:** Simulates fuel price fluctuations (-20% to +60%) with airline pass-through elasticity to model consumer burden in ■ Billions.
- **HHI Antitrust Route Monitor:** Computes Herfindahl-Hirschman Index (HHI) for every route to flag monopoly pricing risks ( $HHI > 2500$ ).

# Slide 3: Technical Approach & Architecture

## 1. Technology Stack & Frameworks

- **Frontend Client:** React 19, TypeScript, Vite, Plotly.js for interactive analytics, Lucide Icons, Modern Vanilla CSS Design System.
- **Backend API:** FastAPI (Python 3.10+), Asynchronous REST microservices, Uvicorn ASGI server with sub-50ms query latency.
- **Statistical Computing:** Python (pandas, numpy, scipy) executing IQR sanitization, median pricing, and Laspeyres index aggregation.
- **Data Persistence:** SQLAlchemy ORM with multi-table relational schema (raw\_fares, representative\_fares, route\_weights, apix\_index) — PostgreSQL production-ready (SQLite portable mode).
- **Orchestration:** Apache Airflow DAG architecture for daily scraping, index computation, and quarterly weight rebalancing.

## 2. End-to-End System Pipeline

### Step 1: Data Ingestion & Deduplication

- Scrapes OTA & airline GDS platforms across 80 routes × 5 horizons.
- Deduplicates fare records based on origin, destination, airline, flight number, and date.

### Step 2: Statistical IQR Outlier Filtration

- Calculates Q1, Q3, and IQR per route/horizon; removes noise & flash anomalies.

### Step 3: Representative Median Computation

- Extracts robust median price  $P_{r,t}$  across clean flight distributions.

### Step 4: Modified Laspeyres Aggregation

- Applies DGCA passenger weights:  $APIx_t = \sum [w_r \times (P_{r,t} / P_{r,0})] \times 100$

# Slide 4: Feasibility, Viability & Risk Analysis

## 1. Multi-Dimensional Feasibility Analysis

- **Operational Feasibility:** Readily combines public DGCA Form A/B city-pair passenger statistics with automated fare ingestion across 20 domestic hubs.
- **Technical Feasibility:** Stateless, lightweight architecture running in real-time with responsive UI and sub-50ms API response latency.
- **Economic Viability:** Zero expensive commercial API dependencies; built 100% on open-source, sovereign, and scalable technology.

## 2. Potential Challenges & Implemented Mitigation Strategies

| Potential Challenge                   | Operational Impact                                                    | Mitigation Strategy Implemented                                                                                                      |
|---------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Extreme Fare Volatility & Flash Sales | Distorts median index with temporary promotional or predatory spikes. | <b>IQR Outlier Trimming:</b> Automatically bounds fares between [Q1 - 1.5·IQR, Q3 + 1.5·IQR] before index calculation.               |
| Seasonal Traffic Pattern Shifts       | Fixed route weights become obsolete over time (e.g. holiday routes).  | <b>Quarterly Weight Rebalancing:</b> Dynamic update engine recalculates route weights ( $w_r$ ) each quarter via DGCA Form A/B data. |
| Anti-Scraping / Portal Rate Limits    | Scraping disruptions leading to incomplete daily fare baskets.        | <b>Distributed Fetching &amp; Robust Fallbacks:</b> Staggered asynchronous requests and persistent SQLite/PostgreSQL caching.        |
| Sovereignty & Territorial Accuracy    | Map rendering errors violating national cartographic guidelines.      | <b>Strict Sol Compliance:</b> Map views rigorously follow Survey of India guidelines (100% integral Indian territory).               |

# Slide 5: Impact and Socio-Economic Benefits

## 1. Direct Stakeholder Impact

- **Ministry of Civil Aviation & DGCA:** Provides an objective, reproducible metric to establish empirical surge thresholds, evaluate fare caps (CAP 3.4), and formulate aviation policies.
- **Competition Commission of India (CCI):** Route-level HHI visibility enables proactive monitoring of airline cartelization and monopoly exploitation on high-density routes.
- **Indian Consumers (Air Travelers):** Complete transparency on fair baseline market fares, surge patterns across lead times, and data-backed optimal booking windows.
- **Airlines & Industry Operators:** Transparent competitive benchmarks without distorting market-driven revenue management.

## 2. Broad Socio-Economic Benefits

- **Macroeconomic Inflation Tracking:** Integrates seamlessly with MOSPI CPI transport sub-indices for inflation forecasting.
- **Regional Connectivity (UDAN Support):** Uncovers price disparities between Tier-1 metro trunk routes and Tier-2/3 regional routes, supporting equitable air connectivity.
- **Consumer Welfare Protection:** Quantifies total national consumer financial impact under fuel shocks or peak festival demand.
- **Sovereign Data Asset:** Establishes India's first standardized, independent air transport price tracking infrastructure.

# Slide 6: Research, Citations & Standards

## 1. Regulatory Data Sources & Methodological Standards

- **Directorate General of Civil Aviation (DGCA), Govt of India:** Monthly Domestic Air Traffic Statistics, City-Pair Passenger Volume Reports (Form A & B) & On-Time Performance Reports (OTP) — *dgca.gov.in*
- **Ministry of Statistics & Programme Implementation (MOSPI):** Consumer Price Index (CPI) Manual, Laspeyres Index Weighting Principles & Methodological Guidelines — *mospi.gov.in*
- **Competition Commission of India (CCI):** Market Concentration Assessment Standards & Herfindahl-Hirschman Index (HHI) Regulatory Frameworks.
- **Ministry of Petroleum & Natural Gas (MoPNG):** Aviation Turbine Fuel (ATF) Pricing Trends, Tax Structures & Airline Operating Cost Breakdowns (~35-40% fuel cost share).
- **International Civil Aviation Organization (ICAO):** Doc 9626 — Manual on the Regulation of International Air Transport: Tariff Monitoring Guidelines & Airfare Indices.

## 2. Project Repository & Verification

- **Live Evaluation URL:** <http://localhost:5173> (React Frontend) | <http://localhost:8000/docs> (FastAPI Engine)
- **GitHub Repository:** *shreejit106/AIR-FARE-PRICE-INDEX*